



PIERRE-FRANCOIS ROUX

EPIGENETICIST & IN SILICO BIOLOGIST

INFO

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SOCIAL

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REFERENCE

Dr Laurent Le Cam
IRCM - Inserm U1194
laurent.lecam@inserm.fr

Pr Anne Dejean
Institut Pasteur
adejean@pasteur.fr

Pr Sandrine Lagarrigue
Institut Agro Rennes-Angers
slagarrigue@agrocampus-ouest.fr

WORK EXPERIENCE

IRCM - Inserm U1194

Molecular Oncogenesis
PI: Dr Laurent Le Cam
Montpellier, France
2021-Present

Research scientist

p53-dependent epigenetic and metabolic reprogramming during senescence and aging

Study of E4F1-mediated acetyl-CoA synthesis, chromatin remodeling and senescence programs using ATAC-seq, CUT&RUN, RNA-seq, metabolomics, cellular models and in vivo aging models.

Johnson & Johnson

Upstream Skin Research & Innovation
PI: Dr G. Stamatias & Dr T. Oddos
Val-de-Reuil, France
2019-2021

Research scientist

Integrative biology for skin research and innovation

Pre-clinical and clinical research on skin biology based on multi-omics cohorts integrating microbiome, metabolome, transcriptome and phenome data.

Institut Pasteur

Nuclear Organization and Oncogenesis
PI: Dr Anne Dejean & Dr Oliver Bischof
Paris, France
2015-2019

Postdoctoral researcher

Temporal profiling of transcriptome and epigenome organization during cellular senescence

Multi-scale study of senescence in human fibroblasts using ATAC-seq, ChIP-seq, transcriptomics, metabolomics, proteomics and network inference.

INRAe

Genetics and Genomics
PI: Pr S. Lagarrigue & Dr O. Demeure
Rennes, France
2011-2014

PhD

Genome-scale SNP detection for complex traits and editome characterization

Integrative analysis of RNA-seq and genome resequencing data for QTL fine mapping, cis-eQTL analysis, allele-specific expression and RNA editing discovery.

IRSET

Cellular Signaling & Response
PI: Dr Corine Martin-Chouly
Rennes, France
2010

MEng trainee

Lipid metabolism in macrophages during inflammation in cystic fibrosis

Characterization of lipidome and cytokine responses in primary macrophages during inflammatory and infectious contexts.

EDUCATION

Agrocampus Ouest & Universite Rennes 1

2009-2011

MEng in agricultural engineering

MSc in cell and molecular biology

Universite Lille 1

2006-2009

MSc in marine biology and biological oceanography

BSc in biology and ecology

Lycee Albert Chatelet

2004-2006

French CPGE BCPST-Veto

SKILLS - TOOLS - LANGUAGES

Epigenomics

Genomics

Genetics

Aging

Cellular senescence

Systems biology

Cancer biology

Skin biology

NGS

R ●●●●●●●●●●○

Biostatistics ●●●●●●●●●●○

Machine learning ●●●●●●●●●●○

UNIX ●●●●●●●●●●○

Bench work ●●●●●●●●●●○

French
NATIVE

English
FLUENT



PROFILE

My research addresses how metabolic states shape gene regulation, chromatin organization and cell fate decisions in aging, cancer and skin biology.

METHODS

Functional genomics

RNA-seq, ATAC-seq, ChIP-seq,
CUT&RUN, MNase-seq

Integrative omics

Metabolomics, microbiome profiling,
transcriptomics, epigenomics

Computational biology

R/Bioconductor, UNIX, reproducible
workflows, network inference

TOPICS

- Cellular senescence
- Aging
- p53/E4F1 metabolism
- Chromatin regulation
- Skin microbiome
- Quantitative genetics
- RNA editing

RESEARCH

C O N T R I B U T I O N S - P U B L I C A T I O N S - F U N D I N G

RESEARCH CONTRIBUTIONS

p53/E4F1, metabolism and senescence

Dissection of links between p53/pRB signaling, pyruvate metabolism, acetyl-CoA production and chromatin regulation during senescence.

ATAC-seq as an integrative assay

Development of bulk ATAC-seq as a dual genetic and epigenetic assay capturing accessibility, variants, CNVs, mitochondrial features and telomere-associated signals.

Systems view of senescence

Temporal multi-omic analyses identifying enhancer remodeling and AP-1-centered regulatory circuits as determinants of oncogene-induced senescence.

Skin microbiome and metabolome

Integration of microbiome and metabolome profiles to identify microbe-metabolite clusters relevant to skin physiology and early-life maturation.

Quantitative genetics

Combination of QTL mapping, WGS, selection signatures, variant annotation and eQTL analyses to prioritize causal genes for adiposity regulation.

RNA editing

Stringent WGS/RNA-seq strategies to distinguish genuine RNA editing events from sequencing, mapping and reference artifacts.

PUBLICATION RECORD

24 original articles

Including 7 first or co-first author publications and 1 last/corresponding author publication.

Reviews and outreach

2 review articles, 1 didactic article and 1 radio communication.

Valorization

4 patents in skin biology, microbiome/metabolome biomarkers and non-invasive molecular profiling.

SELECTED PUBLICATIONS

Beyond chromatin accessibility: bulk ATAC-seq as an integrative assay to portray genomes and epigenomes.

Toumi I., Kham C., Stuani L., Le Cam L., Roux P.F. NAR Genomics and Bioinformatics, 2026. Last and corresponding author.

E4F1 coordinates pyruvate metabolism and the activity of the elongator complex to ensure translation fidelity during brain development.

Di Michele M., Attina A., Roux P.F. et al. Nature Communications, 2025. Co-second author.

Skin maturation from birth to 10 years of age: structure, function, composition and microbiome.

Stamatas G.N., Roux P.F. et al. Experimental Dermatology, 2023. Co-first author.

Escape from oncogene-induced senescence is controlled by POU2F2 and memorized by chromatin scars.

Martinez-Zamudio R.Y. et al. Cell Genomics, 2023.

3-Deazaadenosine alleviates senescence to promote cellular fitness and cell therapy efficiency in mice.

Guerrero A. et al. Nature Aging, 2022.

Deciphering the role of skin surface microbiome on skin health: an integrative multi-omics approach reveals three microbial/metabolic clusters.

Roux P.F., Oddos T., Stamatas G. Journal of Investigative Dermatology, 2021. First author.

AP-1 imprints a reversible transcriptional programme of senescent cells.

Martinez-Zamudio R.I., Roux P.F. et al. Nature Cell Biology, 2020. Co-first author.

Necroptosis microenvironment determines lineage commitment in liver cancer.

Heinzmann F. et al. Nature, 2018.

Combined QTL and selective sweep mappings with coding SNP annotation and cis-eQTL analysis revealed PARK2 and JAG2 as candidate genes for adiposity regulation.

Roux P.F. et al. G3, 2015. First author.

FUNDING

ANR 2026 - submitted

Metabolism-epigenome interactions during cellular senescence and aging.

Canceropole GSO 2026 - submitted

Integrative tumor profiling by ATAC-seq for actionable signatures.

ANR 2019 - acquired

S-ENCODE project on deciphering the senescence code to improve healthy lifespan.



TEACHING

Since 2025

SIRIC Montpellier Cancer - ATAC-seq
practical training

Since 2021

Universite de Montpellier - Master 2
Biology-Health

Since 2014

Institut Agro Rennes-Angers - Omics,
UNIX/R and NGS training

QUALIFICATION

CNU 64

Biochemistry and molecular biology

CNU 65

Cell biology

TRAINING

2022

Animal experimentation training,
designer level, BioCampus Montpellier

ACTIVITIES

T E A C H I N G - S U P E R V I S I O N - S E R V I C E

TEACHING

ATAC-seq data analysis in practice

SIRIC Montpellier Cancer, since 2025. Practical training for cancer
researchers and clinicians.

Cellular senescence and aging

Universite de Montpellier, Master 2 Biology-Health, since 2021. Molecular
basis of senescence, p53 biology, metabolism and aging-related
pathophysiology.

RNA-seq and epigenomic data analysis

Institut Agro Rennes-Angers / Agrocampus Ouest, since 2014. UNIX,
R/Bioconductor, RNA-seq, ChIP-seq, ATAC-seq and genomic data analysis.

Cancer epigenomics

Universite Paris Diderot. Epigenomics for cancer research and clinic for MD
students, MSc students and health professionals.

ALUMNI AND SUPERVISION

Former trainees

- Chaimae Kam, Master 2 Structural Biology & Genomics, 2025.
- Alizee Lanon, Master 2 Bioinformatics and Biostatistics, 2024.
- Arnaud Ferreira, Master 2 Epigenetics and Cellular Biology, 2023.
- Safiya Atiya, Master 2 AMI2B, 2023.
- Islem Tourmi, Master 2 Structural Biology & Genomics, 2023.
- Pablo Garcia Idieder, Master 2 Cancer Biology, 2022.
- Lisa Salhi, Master 2 Bioinformatics, 2019.
- Camille Pere, Master 1 Cellular and Molecular Biology, 2014.
- Emile Richard, Master 2 Bioinformatics & Genomics, 2013.
- Allan Jincq, Master 1 Cellular and Molecular Biology, 2012.

Doctoral supervision and committees

- Arnaud Ferreira, PhD student, IRCM - Inserm U1194, 2023-2026. Impact
of aging and cellular senescence on metabolic adaptability.
- Alexis Hucteau, PhD thesis committee, Centre de Recherches en
Cancerologie de Toulouse, 2021-2024.

COMMUNICATIONS AND SERVICE

Invited seminars

Systems views of senescence and metabolic reprogramming at TOXALIM,
IRCM and the London Institute of Medical Science - Imperial College.

Conference talks

International Cell Senescence Association, Plant and Animal Genetics,
European Federation of Animal Science and European Symposium on
Poultry Genetics.

Poster communications

Euro Geroscience, International Cell Senescence Association, AVIESAN
modeling workshop, American Society of Human Genetics and EAAP.

Service

Editorial and peer-review activities, scientific animation, collaborative
networks, mentoring and professional equality actions.

PATENTS

- Biomarkers predictive of atopic dermatitis, JCO6146USPSP1, pending, 2022.
- Use of microbiome and metabolome clusters to evaluate skin health, US20220142560A1, published, 2021.
- Use of biomarkers to evaluate the efficacy of a composition reducing the effects of cancer therapeutics on skin, US20220057383A1, published, 2021.
- Non-invasive method to evaluate surface skin miRNome and transcriptome, EP21306470.2, 2021.

RESEARCH PROFILE

I combine experimental molecular biology, functional genomics, multi-omics integration and computational biology to understand how
metabolic states translate into epigenetic and transcriptional programs that govern cell fate decisions in aging, cancer and skin biology.